

Keyush Shah

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EDUCATION

University of Pennsylvania (School of Engineering and Applied Science)

Philadelphia, PA

Master of Science in Engineering (MSE) in Data Science; GPA 3.9/4.0

Aug 2023 – May 2025

Coursework: Machine Learning, Databases, Computer Systems, Deep Learning, Generative Modeling, NLP, Machine Perception

EXPERIENCE

Machine Learning Engineer, [Omnicom](#) (NYC, New York)

Aug 2025 - Present

▪ Synthetic Audiences:

1. Architected a RAG-augmented product using **Next.js + FastAPI on Vercel/Railway** enabling pharma clients to simulate patient focus groups - interviewing AI-grounded personas & **reducing 2-4 weeks of research to minutes**.
 2. Researched and prototyped a statistical Gaussian traversal inference-time persona generator as a clustering-free alternative for data-scarce therapeutic domains – unlocking ad-hoc research via a **sub-second inference call**.
- Led end-to-end migration of a multi-component ML Pipeline from AzureML to Databricks by re-architecting LLM Integration, data storage and maintaining orchestration in Spark Job DAGs unifying pipeline governance.
 - Developed CPU-only **BERTopic** pipeline, replacing the GPU compute node & fetching same output at **~90% lower cost**.
 - Reduced CI/CD pipelines runtime by **12x** by migrating from Pipenv to **UV package manager**, configuring private package indexes and benchmarking Docker containers - packages installation slashed to 12s across backend services.
 - Centralized embedding generation across Databricks pipelines by replacing OpenAI SDK instantiation with endpoint wrapper reducing **~1000 per-run API calls by 10x** through request batching & eliminating per-call auth overhead.

AI Researcher, [Penn Medicine - Computational Social Listening Lab](#) (PA, USA)

Dec 2024 – Present

IH Risk Model [[CODE LINK](#)]

- Reduced **model tuning time by 10x** by developing an **AutoML** pipeline using sklearn for **hernia** prediction (**recall: 0.9**)
- Deployed GPT-4 few-shot pipeline on 10k+ redacted surgical notes, **reducing 30% noise vs. BERT** for feature labeling.
- Built a fault tolerant multi-agent orchestration system using Claude Agent SDK featuring **LLM-as-judge** observability, automated retry logic and **FastAPI + WebSocket web interface** slashing pipeline runtime from **45 to 10 min**.
- Integrated **vLLM** & **LangChain** for scalable local LLM inference deployed on A100 GPU and **modularized the backend** (OpenAI, Mistral) using **parallelized batch inference** and **memory-aware chunking** across variable length notes.

Data Science Intern (Full-time), [Universal Media](#) (PA, USA)

May 2024 – Aug 2024

- Led the development of **3+ data pipelines** using Azure Data Factory, facilitating seamless ingesting into Azure
- Developed python scripts for data transformation, stored them in Blob storage and executed them via batch activity in ADF.
- Drove product insights by building **Mixed Media time series models in Azure Synapse**, analyzing marketing channel impacts on media diversity metrics. Built **Power BI** dashboards to deliver actionable insights for optimizing client strategies.

Assistant Manager (Full-time), [IIFL Finance Ltd](#)

Apr 2022 – July 2023

- Analyzed ETL process failures and created **10+** paginated reports in SSIS to help the management track 1000+ branches.
- **Digital Adoption:** Led a product-focused initiative to identify digitally savvy customers by engineering features and building ADF pipelines to track campaign behavior. Trained and deployed a Random Forest model (**with a 90% accuracy**) in Azure ML Studio; **exposed it as a REST endpoint consumed by marketing campaigns**, driving digital **disbursal adoption by 50%**.

SELECTED PROJECTS

▪ Particle (2024):

- Built a context-aware chatbot for product and order support using FastAPI, LangChain, and Pinecone, implementing **RAG for product queries and tool-based logic with LLM fallback** for transactions.
- Engineered multi-turn memory and entity tracking, **enabling >95% accurate follow-up handling**. Automated ingestion of synthetic data, & delivered **React + Vite frontend, improving support efficiency by 40%** and user satisfaction. [\[Link\]](#)

- **Ride Duration Prediction (2025):** Developed a production-ready ML pipeline to predict NYC taxi ride durations, using **Airflow** for orchestration and **MLflow** for experiment tracking. Achieved **~30% RMSE reduction** via automated hyperparameter tuning & designed modular, reproducible workflows to simulate real-world deployment as a web service via **Flask**. [\[Link\]](#)

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, JavaScript, Typescript, R programming

ML Libraries/Frameworks: PyTorch, scikit-learn, XGBoost, LightGBM, HuggingFace, spaCy, OpenCV, MLflow, A/B Testing

Databases/Web Frameworks: MySQL, PostgreSQL, SSMS, MongoDB, Neo4j, React, NodeJS, Django, Flask, FastAPI

Cloud/Big Data Orchestration: AWS (S3, Glue), Azure (Data Factory, Synapse), GCP (BigQuery), Kafka, Airflow, DBT, PySpark

Tools/DevOps: DataBricks, Apache Spark, Docker, Jenkins, Git, Kubernetes, pytest, CI/CD